Fiscal 2025 R&D Subsidy (Overseas) Program Application Form

25/08/2025

I. Theme of R&D Application (Within 20 words)

Early Detection of Alzheimer’s Disease and Mild Cognitive Impairment in Vietnamese older adults through speech analysis

II. Application Amount

Yen : 1,747,652

III. Applicant

Name	Ha Thi Thanh Huong				
Education level	PhD	Age	35	Gender	Female
Organization	International University-Vietnam National University Ho Chi Minh City	Position	Head of Department		
URL	https://hcmiu.edu.vn/en/				
Location	VNUHCM Township, Quarter 33, Linh Xuan Ward, Ho Chi Minh City				
E-mail	htthuong@hcmiu.edu.vn	Tel.	(+84) 938939445		
Affiliated academic societies	IBRO				
Summary of Applicant (Please state post-educational experience)	After graduating with a PhD degree from Stanford University, I decided to return to Vietnam in 2018. Upon coming back to Vietnam, I established the Brain Health Lab and have served as lecturer and department chair of Tissue Engineering and Regenerative Medicine at International University-Vietnam National University Ho Chi Minh City for over 6 years. My research focuses on addressing mental health issues and aging disorders to improve the quality of life. My team conducted an experiment that applies AI and a brain-computer interface system to support the rehabilitation of stroke patients. Additionally, we investigated AI potential in detecting brain tumors and migraines. Furthermore, I am interested in exploring the relationship between stress and emotions as indicated by changes in EEG signals, along with evaluating the effectiveness of techniques like alpha music and mindfulness-based stress reduction for alleviating stress. Besides, I am also interested in enhancing diagnosis and interventions for dementia at an early stage. In terms of the diagnosis aspect, I am leading the early diagnosis of Alzheimer’s Disease, focusing on detecting p-Tau217 and cell-free RNA detection. For studies related to AI, I am developing an easy-to-use AI system to detect early signs of Alzheimer’s disease in clinical settings or the biological characteristics of heterogeneity in the Mild Cognitive Impairment (MCI) population. When it comes to the intervention aspect, my research team developed BrainTrain for MCI patients to hinder their conversion to dementia and evaluate the efficacy of EEG signals in predicting post-intervention effectiveness. Our application has reached over 150 patients, providing accessible home-based dementia intervention for underserved communities. I am leading projects to build the first Vietnamese databases of neuroscience-related data, such as brain MRI and EEG signals to enhance AI models and improve diagnosis. Finally, my goal is to create innovative, highly accurate, and low-cost products that can have a tangible impact on the brain health and mental well-being of those in my community and beyond.				

IV. Recommender

I hereby recommend this candidate to TERUMO LIFE SCIENCE FOUNDATION’s R&D Subsidies (Overseas) Program. I am the head of the research institution to which the applicant belongs. I recommend only one application.

Name (Handwritten Signature)	Associate Professor Le Van Thang		
Organization	International University, Vietnam National University Ho Chi Minh City	Position	President
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